

Data

speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
	$\left(\frac{1}{4\pi\epsilon_0}\right) = 8.99 \times 10^9 \text{ m F}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T / \text{K} = T / ^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.

velocity of particle in s.h.m.

electric current

resistors in series

resistors in parallel

capacitors in series

capacitors in parallel

energy in a capacitor

charging a capacitor

discharging a capacitor

RC time constant

electric potential

alternating current / voltage

magnetic flux density due to a long straight wire

magnetic flux density due to a flat circular coil

magnetic flux density due to a long solenoid

energy states for quantum particle in a box

radioactive decay

radioactive decay constant

$$x = x_0 \sin \omega t$$

$$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$$

$$I = nAvq$$

$$R = R_1 + R_2 + \dots$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

$$C = C_1 + C_2 + \dots$$

$$U = \frac{1}{2}QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2}CV^2$$

$$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$$

$$Q = Q_0 e^{-\frac{t}{\tau}}$$

$$\tau = RC$$

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

$$x = x_0 \sin \omega t$$

$$B = \frac{\mu_0 I}{2\pi d}$$

$$B = \frac{\mu_0 NI}{2r}$$

$$B = \mu_0 nI$$

$$E_n = \frac{h^2}{8mL^2} n^2$$

$$x = x_0 e^{-\lambda t}$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$